

Continued Fractions and Ramanujan Operators for QML Deep Brain Stimulation Optimization

An Extension for MS and Alexander's Disease

Abstract:

We extend our prior Quantum Machine Learning (QML) and finite mathematical framework for neuro-electrodynamic ablation by introducing Ramanujan operators and infinite continued fraction (CF) expansions. These operators bound the error gradients in plaque density mitigation for Multiple Sclerosis (MS) more strictly than discrete finite schemas. Additionally, we provide an Addendum analyzing neural recovery and Rosenthal fiber ablation in Alexander's Disease utilizing a novel CF-bounded model, yielding highly asymmetric, infinitely convergent recovery curves.

1. MS Paradigm Improvement via Ramanujan Operators

In our MS plaque mitigation paradigm, we substitute standard finite difference QML cost functions with a Ramanujan modular expansion. Plaque density $P(t)$ is now modeled as a continuously converging fraction evaluating the field potential E :

$$P(t) = \frac{1}{1 + \frac{\alpha E(t)}{1 + \frac{\beta E(t)^2}{1 + \dots}}}$$

This Ramanujan operator enforces a bounded saturation limit on mitigation, preventing overshoot in stochastic QML evaluations.

2. Addendum: Continued Fractions for Alexander's Disease

For Rosenthal fiber ablation, we map the structural degradation of the fibers $A(t)$ along Neural Recovery $R(t)$ forming a coupled system of continued fractions. The ablation rate

$$A(t) = A_0 - \frac{\kappa}{\delta + \frac{t}{1 + \frac{t^2}{2 + \dots}}}$$

Here, δ reflects the impedance resistance of mutant GFAP aggregates. This ensures the neural recovery metrics dynamically converge exactly as Rosenthal fibers degenerate safeguarding surrounding white matter astrocytes.